

HOMEWORK ASSIGNMENT 2 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released at 8:00am on Thursday, September 3 in time for the corresponding quiz on **Wednesday September 9** (note that Monday September 7 is a holiday!)
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1. **Commute Times.** Vanessa wants to determine the average commute time (time taken from place of residence to place of work/school) across all California residents. To collect her data, she goes to the Cal Poly registrar's office, obtains the emails of all currently-enrolled Cal Poly students, and sends a survey to their email addresses asking what their commute times are. 15 Cal Poly students respond to her survey; these 15 students have an average commute time of 7.23 minutes.

- (a) Describe an observational unit in the context of this dataset.

Solution: An observational unit is **one person**.

- (b) Define the population of interest.

Solution: The population is **the set of California residents**.

- (c) Define the parameter of interest. Be sure to use proper mathematical notation!

Solution: The parameter of interest is μ , the mean commute time across all California residents.

- (d) What is the sampling frame?

Solution: The sampling frame is, by definition, the set of all units that could have been included in Vanessa's sample. Because she is collecting her data by emailing currently-registered Cal Poly students, her sampling frame is **the set of all Cal Poly Students**.

- (e) Did Vanessa obtain a simple random sample? Why or why not?

Solution: No, because the sampling frame is much smaller than the population and not all samples of size 15 are equally likely (which they would be if the sample Vanessa obtained were truly a simple random sample).

- (f) Do you think Vanessa's sampling method is biased or unbiased? If you think it is biased, do you think it has a tendency to underestimate or overestimate the true value of the parameter of interest? **Hint:** recall that residents of large, sprawling cities like San Francisco and Los Angeles are also California residents.

Solution: As the hint points out, residents in large cities like San Francisco or Los Angeles are also included in the population. However, these residents would likely have much *longer* commute

times than residents of San Luis Obispo (which is a fairly small town). Furthermore, it's not far-fetched to surmise that college students tend to live closer to their universities and places of work. So, all in all, the samples Vanessa obtains (by sampling only Cal Poly students) are likely to give her a underinflated sense of how long the true average commute for a California resident is. That is, her sampling method is likely to be **biased, with a tendency to underestimate**.

- (g) How generalizable do you think Vanessa's findings will be? Why?

Solution: Vanessa will have limited generalizability, because of the bias present in her sampling method. The samples she obtains will not necessarily be representative of the population, inhibiting her ability to generalize her findings.

2. **Going Gaga for Lady Gaga.** What is the average length (in minutes) of a Lady Gaga song? To answer this question, Morenike went online to repository that maintains a list of all Lady Gaga songs, and assigned a unique number to each Lady Gaga song. She then used a random number generator to generate 20 numbers, and looked up the 20 Lady Gaga songs corresponding to these numbers. These 20 Lady Gaga songs had an average runtime of 3.56 minutes.

- (a) Describe an observational unit in the context of this dataset.

Solution: One observational unit is **one song**.

- (b) Define the population of interest.

Solution: The population is the set of **all Lady Gaga songs**.

- (c) What is the sampling frame?

Solution: The sampling frame is the set of all Lady Gaga songs, since Morenike theoretically could have sampled any Lady Gaga song (we are told that the *Wikipedia* page she is using lists all Lady Gaga songs).

- (d) Did Morenike obtain a simple random sample? Why or why not?

Solution: Yes. First, the sampling frame is identical to the population; additionally, every unit in the population was equally likely to have been included in Morenike's sample (since she used a random number generator).

- (e) How generalizable do you think Morenike findings will be? Why?

Solution: Given that Morenike took a simple random sample, her sample is likely to be quite representative of the population; therefore, her findings are likely to be **quite generalizable**.

- (f) Suppose Morenike finds out that the repository she was using actually doesn't contain *all* Lady Gaga songs, and some tracks were not listed. What type of bias does this introduce into Morenike's sample? Make sure you've read all of Lecture Activity 3!

Solution: This is a type of **non-response** bias.

3. **Restaurant Ratings.** Two friends, Agnieszka and Bianca are having a conversation about restaurant ratings. In particular, they are looking at the online reviews for a local restaurant. Agnieszka argues that the average rating listed online is probably lower than the “true” rating, given that people tend to be more inclined to write reviews when they have a particularly negative experience. Bianca argues that, though that may be true, the restaurant has thousands of ratings and therefore the average rating listed online is probably pretty accurate.

(a) Rephrase Agnieszka’s argument in the language of sampling discussed in Lecture Activity 3. In particular, make note of:

- The parameter of interest
- An observational unit in the context of the problem
- The population
- The sampling frame
- The sampling method
- Whether Agnieszka is claiming their sampling method to be biased or unbiased, and if it is biased whether she’s claiming the method has a tendency to over- or underestimate.

Solution: The quantity Agnieszka and Bianca are trying to estimate is μ , the average rating across all diners who’ve dined at this particular restaurant. The observational units are therefore diners, and the one variable in this dataset is the rating each diner gave to the restaurant. The sampling frame is the set of all diners that posted their ratings online; Agnieszka and Bianca’s sampling method is to simply collect all units in the sampling frame. Agnieszka is arguing that this sampling method is **biased**, with a tendency to **underestimate**.

(b) Is Bianca’s argument correct, or is she making a mistake? If she’s making a mistake, what mistake is she making? (**Hint:** according to what we saw in Lecture Activity 3, there is a correct answer here!)

Solution: Bianca is essentially arguing that a large enough sample size can counteract the effects of bias. As we saw in Lecture Activity 3, this is not necessarily the case; **Bianca’s logic is flawed**.

4. **Possible Probabilities.** For each of the following values, determine whether it is a valid probability value or not. For those that are not valid probability values, explain why they are not valid probability values.

(a) 0.00

Solution: 0.00 is a valid probability value.

(b) 0.8

Solution: 0.8 is a valid probability value.

(c) 1.25

Solution: 1.25 is not a valid probability value since probabilities cannot be greater than 1.

(d) -0.43

Solution: -0.43 is not a valid probability value, since probabilities cannot be negative.

(e) 0.2

Solution: 0.2 is a valid probability value.

5. **Coin Conundrum.** Marissa has four coins, but she's concerned that they may not be fair (i.e. that they may land "heads" more or less often than "tails"). To that end, she enlists the help of her friends and tosses each coin 3000 times and records the relative proportion of heads at each stage. Shown below are the relative frequency diagrams associated with each coin (labeled "A" through "D")

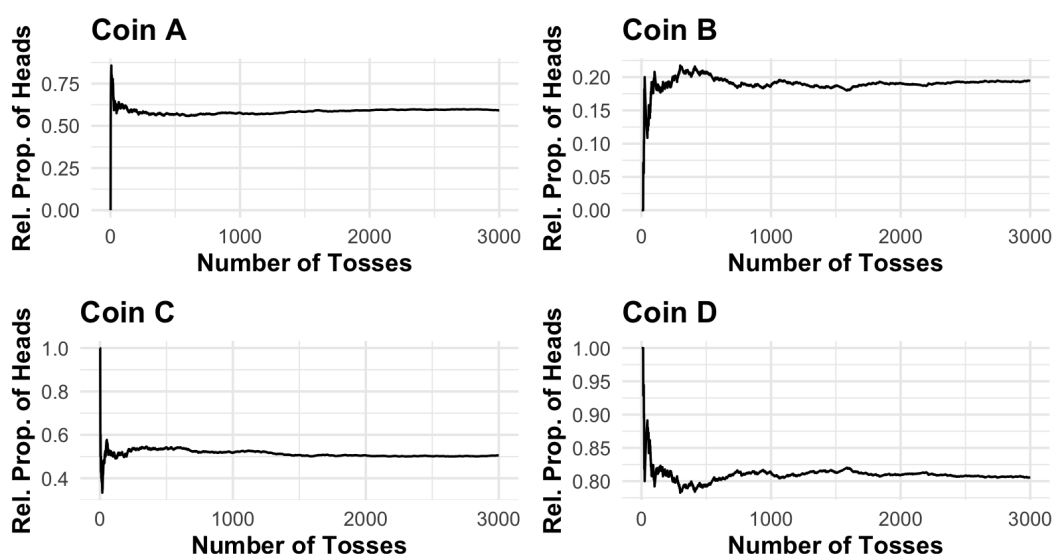


Figure 1: Four relative frequency diagrams.

- (a) What is the probability that Coin A lands "heads?" How can you tell? Is Coin A fair?

Solution: The probability of an event is defined to be the long-run proportion of times it occurs. We see that the long-run proportion of times Coin A lands heads is around 0.6, meaning the probability that Coin A lands heads is $0.6 = 60\%$. Since this is not equal to 50%, Coin A is not fair.

- (b) What is the probability that Coin B lands "heads?" How can you tell? Is Coin A fair?

Solution: The probability of an event is defined to be the long-run proportion of times it occurs. We see that the long-run proportion of times Coin B lands heads is around 0.2, meaning the probability that Coin B lands heads is $0.2 = 20\%$. Since this is not equal to 50%, Coin B is not fair.

- (c) What is the probability that Coin C lands "heads?" How can you tell? Is Coin A fair?

Solution: The probability of an event is defined to be the long-run proportion of times it occurs. We see that the long-run proportion of times Coin C lands heads is around 0.5, meaning the probability that Coin C lands heads is $0.5 = 50\%$. Since this is equal to 50%, **Coin C is fair.**

(d) What is the probability that Coin D lands “heads?” How can you tell? Is Coin A fair?

Solution: The probability of an event is defined to be the long-run proportion of times it occurs. We see that the long-run proportion of times Coin D lands heads is around 0.6, meaning the probability that Coin D lands heads is $0.6 = 60\%$. Since this is not equal to 50%, **Coin D is not fair.**

A Note: Technically speaking, for all parts on this question, we’re only able to get an *approximate* probability each coin lands heads. Recall that, technically, to get the *true* probability, we would need to conduct an infinite number of representations. Think back to the Monty Hall example from Lecture Activity 4, and the relative frequency diagrams you obtained from the applet.

6. **Killer Cereal.** *Killer Cereal*, a brand of breakfast cereal, claims that there is a 1 in 4 chance that a box contains a prize. That is; they claim the probability that a box contains a prize is $\frac{1}{4} = 0.25 = 25\%$. Which of the following are valid interpretations of this probability? Select all that apply.

- ☐ Among the next four boxes of *Killer Cereal* you buy, exactly 1 will contain a prize.
- ☒ **In a very large sample of *Killer Cereal* boxes, we would expect around $\frac{1}{4}$ of them to contain a prize.**
- ☐ In a sample of 80 *Killer Cereal* boxes, around 20 would contain a prize
- ☐ One in every four boxes of *Killer Cereal* will, with absolute certainty, contain a prize
- ☐ One box of *Killer Cereal* is likely to contain a quarter of a prize

Solution: The first statement is too strong: remember that probabilities are not guarantees! The third choice is tempting, but remember that probabilities are *long-run* proportions. (Stated differently, it’s unclear whether a sample size of 80 is large enough to be in the “long-run.” If the statement had read “In a sample of 80 *Killer Cereal* boxes we would **expect** to find *around* 20 that have a prize,” then we might be in business.) The fourth choice is, again, too strong. The fifth statement is incorrect - a box cannot contain a fraction of a prize.

A quick note - in hindsight, the third answer choice is perhaps a little too similar to a correct statement than I would have liked. That is to say - on a quiz, I’ll try and make it more clear which choices are incorrect and which are correct.